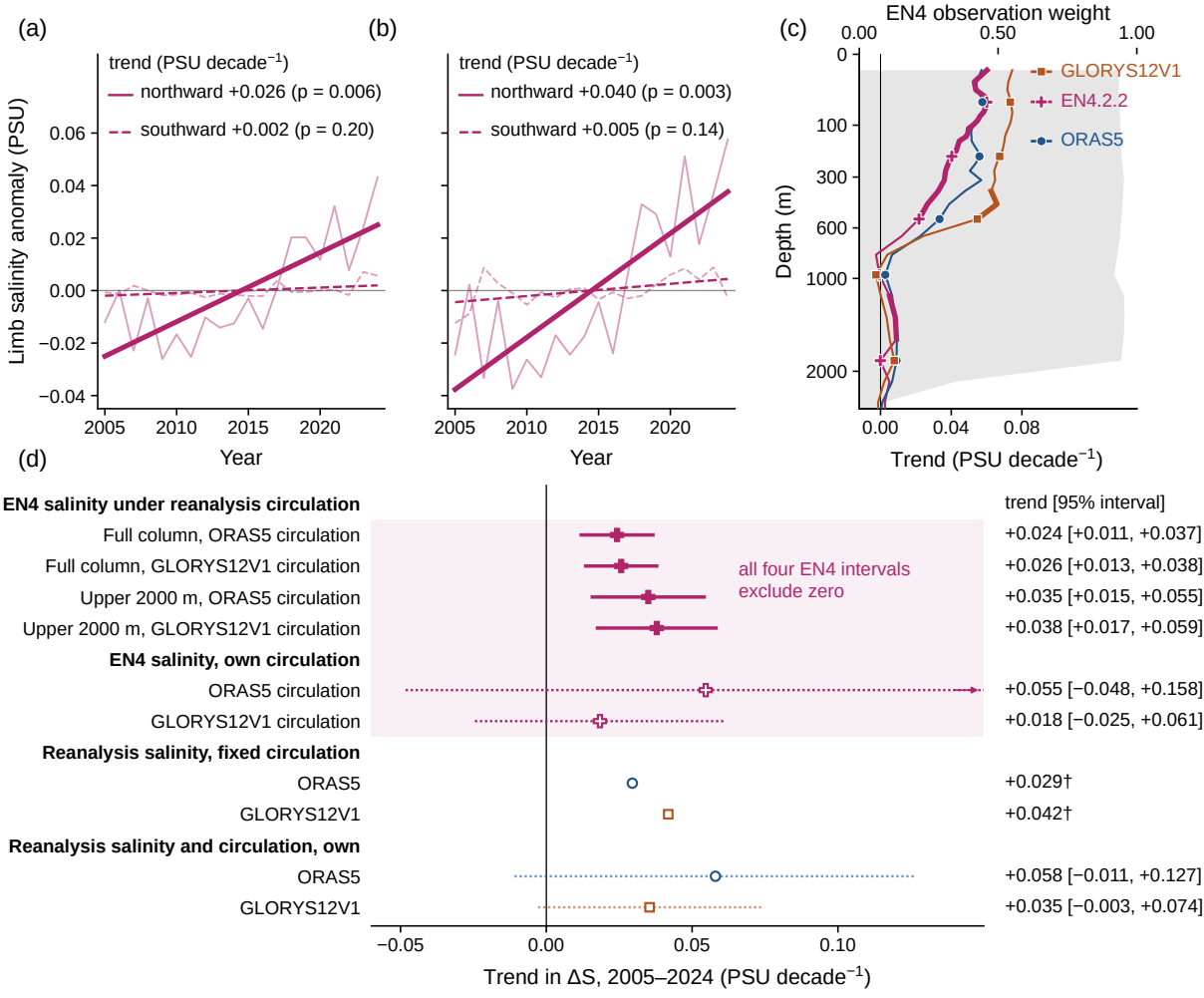
(a)
trend (PSU decade^{-1})
northward +0.026 (p = 0.006)
southward +0.002 (p = 0.20)
Limb salinity anomaly (PSU)
0.06
0.04
0.02
0.00
−0.02
−0.04
2005
2010
2015
2020
Year
(b)
trend (PSU decade^{-1})
northward +0.040 (p = 0.003)
southward +0.005 (p = 0.14)
2005
2010
2015
2020
Year
(c)
EN4 observation weight
0.00
0.50
1.00
GLORYS12V1
EN4.2.2
ORAS5
Depth (m)
0
100
300
600
1000
2000
0.00
0.04
0.08
Trend (PSU decade^{-1})
(d)
EN4 salinity under reanalysis circulation
trend [95% interval]
Full column, ORAS5 circulation
+0.024 [+0.011, +0.037]
Full column, GLORYS12V1 circulation
+0.026 [+0.013, +0.038]
Upper 2000 m, ORAS5 circulation
+0.035 [+0.015, +0.055]
Upper 2000 m, GLORYS12V1 circulation
+0.038 [+0.017, +0.059]
all four EN4 intervals
exclude zero
EN4 salinity, own circulation
ORAS5 circulation
+0.055 [−0.048, +0.158]
GLORYS12V1 circulation
+0.018 [−0.025, +0.061]
Reanalysis salinity, fixed circulation
ORAS5
+0.029†
GLORYS12V1
+0.042†
Reanalysis salinity and circulation, own
ORAS5
+0.058 [−0.011, +0.127]
GLORYS12V1
+0.035 [−0.003, +0.074]
−0.05
0.00
0.05
0.10
Trend in ΔS, 2005–2024 (PSU decade^{-1})